

MAE5005 Computational Fluid Dynamics, Fall 2026

Homework Set 3

Due Wednesday, March 25, 2026 by 23:59, by online submission through *Blackboard*

9. (20 points) Consider the following finite-difference representation for the cross partial derivative,

$$\frac{\partial^2 u(i, j)}{\partial x \partial y} \approx \frac{u_{i+1, j+1} - u_{i-1, j+1} - u_{i+1, j-1} + u_{i-1, j-1}}{4\Delta x \Delta y} \quad (1)$$

- (a) What is the exact expression of the leading-order error term?
 - (b) Now if one applies the above approximation to the location $x = 0$ and $y = 1$, with $\Delta x = \Delta y = 0.1$, for a field $u(x, y) = y^4 \sin x$, compute the exact value of the cross partial derivative and the value from the finite-difference approximation.
 - (c) Confirm that the difference between the exact value and the value from the finite-difference approximation is indeed very similar to the leading-order error term you obtained in (a).
10. (10 points) Consider the following nonlinear differential equation (known as the Bernoulli differential equation)

$$\frac{du(t)}{dt} = F(u, t), \quad \text{with } F(u, t) = \frac{2u}{t} - t^2 u^2. \quad (2)$$

- (a) First, verify that

$$u(t) = \frac{5t^2}{t^5 + C} \quad (3)$$

satisfies the above nonlinear differential equation. Here C is an arbitrary constant.

- (b) With the above solution, compute explicitly

$$\frac{DF}{Dt} = \frac{\partial F}{\partial u} \frac{du(t)}{dt} + \frac{\partial F}{\partial t}, \quad (4)$$

in terms of t . Note that the above notations were used in our lecture.

- (c) Confirm that the result in (b) is identical to d^2u/dt^2 .

Note: I made this problem purposely for you to understand what we used in the lecture notes.

11. (30 points) Consider the 1D transient laminar flow problem again. The problem is formulated by the following partial differential equation

$$\frac{\partial u}{\partial t} = g + \nu \frac{\partial^2 u}{\partial y^2}, \quad -L \leq y \leq L \quad (5)$$

where $u(y, t)$ is the flow velocity, ν is the kinematic viscosity, L is the half channel width. The body force per unit mass is related to the maximum flow velocity u_0 at steady state as $g = 2\nu u_0/L^2$. The initial and boundary conditions are

$$u(y, t = 0) = 0, \quad u(y = \pm L, t) = 0. \quad (6)$$

The central-differencing-in-space and forward-Euler-in-time finite-difference formulation is

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = g + \nu \frac{u_{j-1}^n - 2u_j^n + u_{j+1}^n}{\Delta y^2}, \quad \text{for } j = 2, 3, 4, \dots, N-1, N. \quad (7)$$

where $u_j^n = u(y_j, t_n)$ with $y_j = -L + (j-1)\Delta y$ and $t_n = n\Delta t$. Here $\Delta y = 2L/N$ and Δt is the time step size. Assume $L = 1$, $u_0 = 1$, $\nu = 0.1$. Define $\alpha \equiv \nu\Delta t/\Delta y^2$. In the

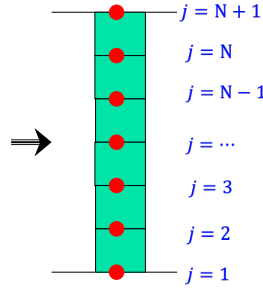


Figure 1: Node points used in the finite-difference algorithm.

recent lecture, we derived the analytical solution for the modified equation

$$\frac{u_m(y, t)}{u_0} = \left(1 - \frac{y^2}{L^2}\right) - \sum_{k=0}^{\infty} \frac{4(-1)^k}{[\pi(k+0.5)]^3} \exp\left[-\frac{(k+0.5)^2\pi^2\nu_m(k)t}{L^2}\right] \cos\left[(k+0.5)\pi\frac{y}{L}\right] \quad (8)$$

which implies that an effective viscosity $\nu_m(k)$ in “the method of separation of variables” solution (Problem 1):

$$\nu_m(k) = \nu \left[1 + \left(\frac{\alpha}{2} - \frac{1}{12}\right) \left(\frac{2(k+0.5)\pi}{N}\right)^2 + \left(\frac{\alpha^2}{3} - \frac{\alpha}{12} + \frac{1}{360}\right) \left(\frac{2(k+0.5)\pi}{N}\right)^4 \right] \quad (9)$$

(a) Develop a code to compare the normalized error predicted, namely,

$$\left| \frac{u_m(y, t) - u(y, t)}{u_0} \right| \quad (10)$$

with the normalized error from the actual numerical simulation

$$\left| \frac{u_N(y, t) - u(y, t)}{u_0} \right| \quad (11)$$

for the case of $N = 16$ and $\alpha = 0.32$. Plot and compare the two at the channel center ($y = 0$) as a function of dimensionless time. Here u_N, u_m, u denote the numerical solution, the analytical solution of the modified equation, the analytical solution of the original problem, respectively.

- (b) Using the solution above, predict the normalized errors in the velocity profile and compare with the normalized errors in the numerical solutions for $N = 8, 16, 32$ for a fixed $\alpha = 0.32$, at two dimensionless times $\nu t/L^2 = 0.2$ and 10.0 . Comment on your results.
- (c) Use your code, for the case of $N = 32$, try 4 different α values, $\alpha = 0.50, 0.505, 0.51, 0.52$, to see if the code is stable by plotting the velocity at the centerline ($y=0$) as a function of dimensionless time $\nu t/L^2$ up to $\nu t/L^2 = 2.0$.